Iuliia (Julia) Vitiugova

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Professional Experience

CTO, Szylla, London *Aug 2026 - Present*
AI investigations for high-risk due diligence. Building specialist AI agents that investigate companies and people and return risk reports with every finding traced to its source.

Lead Gemini Ambassador, Google *Apr 2026 - Aug 2026*
Led a 20-person student team running events, masterclasses and hackathons to grow Gemini use across Cambridge. Tested new Gemini products and fed back directly to Google engineers and product leads, proposing new researcher-centred features.

Data Scientist (Contract), Icorp-Med (Health-tech VC), Paris *May 2025 - Oct 2025*
For a portfolio company, worked on the signal-processing pipeline of a CE-marked cardiovascular device used by clinicians, making it 3x faster. Researched and advised on integrating DL-based signal quality control into the existing device and software.

Machine Learning Intern, Yandex, Moscow *Jun 2021 - Sep 2021*
Implemented production machine learning models for search ranking and content classification systems.

Education

University of Cambridge *Oct 2025 - Jun 2026*
MPhil in Math, Physics, and Deep Learning
Department of Applied Mathematics and Theoretical Physics (DAMTP)
MPhil Thesis: Mechanistic Interpretability of Open Source LLMs
Supervisor: Dr. Miles Cranmer

Université Sorbonne Paris *Sep 2024 - May 2025*
M1 in Computer Science

Lomonosov Moscow State University *Sep 2019 - Jun 2023*
BSc in Physics Graduated with Honours
Faculty of Physics, Department of Mathematical Modelling and Computer Science
BSc Thesis: Analysis of Neural Network Weights in Data Processing in Physics
Supervisor: Dr. Sergey Dolenko

Research Experience

Research Assistant *Nov 2025 - Aug 2026*
DAMTP, University of Cambridge
Researched internal representations, circuit-level behaviour and the mathematical structure underlying modern transformer architectures. Built a pipeline that finds which internal circuits drive an LLM's answers, separating cause from correlation with intervention experiments, run on Cambridge's GPU cluster.

Research Assistant *Sep 2020 - Dec 2023*
Skobeltsyn Institute of Nuclear Physics Moscow
Laboratory of Adaptive Methods of Data Processing

Machine Learning and Deep Learning for Sensors Data:

- Developed and trained classifier and non-linear regression models for gas sensing detection (PyTorch, Scikit-learn). Developed advanced signal processing methods for pattern recognition of data obtained from semiconductor sensors in natural air (Feature Selection and Extraction methods).
- Evaluated the performance of models and refined predictions (Calibration methods). Conducted interpretability analysis of neural networks (Weight Analysis, Deep Taylor Decomposition).

Deep Learning for Spectroscopy Data:

- Developed and trained DL models (PyTorch, Scikit-learn).
- Conducted interpretability analysis for neural networks, particularly focusing on Weight Analysis methods for spectroscopy (IR/Raman) databases.

Collaborative Chemistry Research:

- Collaborated with a chemistry group (Faculty of Chemistry, MSU) to apply Chemometrics methods for computing energy characteristics of molecules in liquids (MCR-methods).

Publications

Comparison of Input Feature Selection Methods Based on Neural Network Weight Analysis *2023*
XXV International Scientific and Technical Conference Neuroinformatics-2023
Vitiugova I., Efitorov A., Shiroky V.

Calibration of Gas Sensor Data Classification Models *2023*
International Conference on Exploring Discoveries and Innovations in Overcoming Global Challenges: A Focus on Fundamental and Applied Research
Vitiugova I., Efitorov A., Dolenko S.

Calibration of Gas Sensor Data Classification Models *2023*
International Journal Of Professional Science №7-2023
Vitiugova I., Efitorov A., Dolenko S.

Conferences

XXV International Scientific and Technical Conference Neuroinformatics-2023 *2023*
Moscow, Russia

International Conference on Exploring Discoveries and Innovations in Overcoming Global Challenges: A Focus on Fundamental and Applied Research *2023*
Seattle, USA

Honors & Awards

Winner of Best AI Project *2026*
Google Gemini for Students competition, among students and Gemini ambassadors

Winner of Best AI Product and Research *2026*
University of Cambridge & FactTrace Inaugural Hackathon, created multi-agent system for AI truth verification and conducted research.

Finalist, PayPal Challenge *2026*
Cambridge Policy Hackathon, Cambridge Judge Business School

Winner of Best AI Project *2025*
University of Cambridge & OpenAI Hackathon for developing fully functional AI-powered product

Scholarship from Murray Edwards College *2025*
University of Cambridge, for MPhil studies

Scholarship for International Studies *2025-2026*
Association Solidarity FRANCE, French Government for Cambridge studies

Excellence Scholarship EUR PNGS-M&CS *2024 - 2025*
Université Sorbonne Paris XIII (Issued by Graduate School in Mathematics and Computer Sciences)

Winner of The Best Innovative Research Project *2023*
International Competition of Scientific and Technical Research Projects: "Science in Motion: Discoveries in Engineering, Mathematics and Physics"

Second Degree Diploma, The Best Innovative Research Project *2023*
International Competition: "Innovations in the Field of Information Technology"

Winner of Universiade of the Faculty of Physics *2023*
Graduate Papers Competition in Lomonosov Moscow State University

State Fellow for Special Academic Achievements *2020 - 2023*
Lomonosov Moscow State University (Issued by Russian Government)

Skills

Programming Languages: Python, C++, R, SQL, Java, MATLAB, LaTeX
ML Frameworks: PyTorch, TensorFlow, Scikit-learn
Languages: English (proficient), Russian (native), French (intermediate)